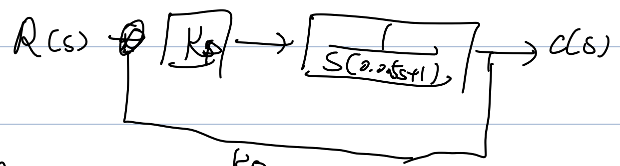
T2.

Q1. b).



$$\frac{C(s)}{R(s)} = \frac{\frac{K_p}{S(0.05s+1)}}{1+\frac{K_p}{S(0.05s+1)}} = \frac{K_p}{0.05S^2+S+K_p} = \frac{20K_p}{S^2+20s+20K_p}$$

$\Delta = 400 - 80K_p \geq 0$

$\Rightarrow K_p \leq 5$

i) $$C(s) = \frac{10}{S(S^2+20s+10)} = \frac{A}{S} + \frac{B}{(S-S_1)} + \frac{C}{S-S_2}$$

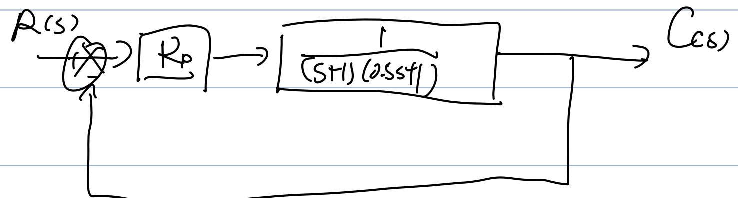
$$S_{1,2} = -10 \pm \sqrt{100-10} = -10 \pm 3\sqrt{10} \quad -19.49 \quad -0.51$$

$$A(S-S_1)(S-S_2) + BS(S-S_2) + CS(S-S_1) = 10$$

$$\Rightarrow (A+B+C)S^2 - \left[A(S_1+S_2) - (BS_2+CS_1)\right]s + S_1S_2A = 10$$

$$\Rightarrow \begin{cases} A+B+C=0 \\ A(S_1+S_2) + BS_2+CS_1 = 0 \\ S_1S_2A = 10 \end{cases} \Rightarrow \begin{cases} A = \frac{10}{S_1S_2} = 1 \\ B = 0.027 \\ C = -1.027 \end{cases}$$

Q2.



$$\frac{C(s)}{R(s)} = \frac{\frac{K_p}{\sim}}{1+\frac{K_p}{\sim}} = \frac{K_p}{0.5S^2+1.5S+1+K_p}$$

$$= \frac{2K_p}{S^2+3S+2(1+K_p)}$$

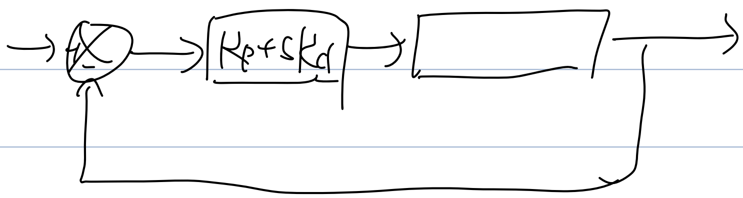
$$C(s) = \frac{2K_p}{s\left[s^2+3s+2(1+K_p)\right]}$$

$$c(\infty) = \lim_{s\to 0} s\,C(s) = \frac{K_p}{1+K_p}$$

$$e(\infty) = \frac{1}{1+K_p} = 1\% \Rightarrow K_p = 99$$

$$\frac{C(s)}{R(s)} = \frac{198}{s^2+3s+200}$$

$$\Rightarrow \begin{cases} 2\zeta\omega_n = 3 \\ \omega_n^2 = 200 \end{cases} \Rightarrow \begin{cases} \omega_n = 10\sqrt{2} \\ \zeta = \dfrac{3}{20\sqrt{2}} \approx 0.106 \end{cases}$$



$$\frac{C(s)}{R(s)} = \frac{2(K_p+sK_d)}{s^2+3s+2(1+K_p+sK_d)}$$

$$= \frac{2K_d s + 200}{s^2+(3+2K_d)s+200}$$

$$\Rightarrow \begin{cases} \omega_n^2 = 200 \\ 2\zeta\omega_n = 3+2K_d \end{cases} \Rightarrow \zeta = \frac{3+2K_d}{20\sqrt{2}} = 0.8$$

$$\Rightarrow K_d = 9.814$$

Q3. a) $G_c(s) = K_p$

$$\frac{\theta(s)}{\theta_R(s)} = \frac{\frac{K_p}{Js^2}}{1+\frac{K_p}{Js^2}} = \frac{\frac{K_p}{J}}{s^2+\frac{K_p}{J}}$$

$$\theta(s) = \frac{1}{s}\cdot\frac{\frac{K_p}{J}}{s^2+\frac{K_p}{J}} = \frac{A}{s} + \frac{Bs+C}{s^2+\frac{K_p}{J}} = \frac{(A+B)s^2+Cs+\frac{AK_p}{J}}{s\left(s^2+\frac{K_p}{J}\right)}$$

$A+B=0 \quad C=0 \quad \frac{AK_p}{J} = \frac{K_p}{J} \Rightarrow A=1 \quad B=-1 \quad C=0$

$$\Rightarrow \theta(s) = \frac{1}{s} - \frac{s}{s^2+\frac{K_p}{J}}$$

$$\omega_n = \sqrt{\frac{K_p}{J}} = \sqrt{0.2} = \frac{1}{\sqrt{5}}$$

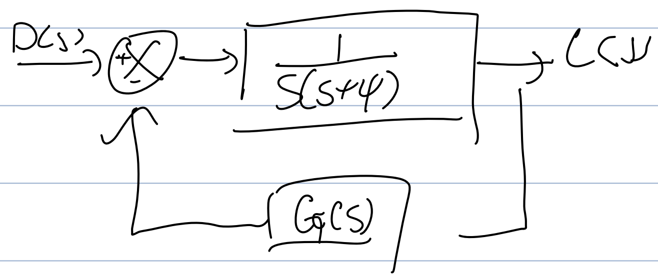
$$T = \frac{2\pi}{\omega_n} = 2\pi\sqrt{5} = 14.05\ s$$

b) $$\frac{\theta(s)}{\theta_R(s)} = \frac{\frac{K_d}{J}s + \frac{K_p}{J}}{s^2+\frac{K_d}{J}s+\frac{K_p}{J}}$$

$$\left(\frac{K_d}{J}\right)^2 > 4K_p$$

$$\Rightarrow K_d > 2\sqrt{K_pJ}$$

1998. a)



$$\Rightarrow \frac{C(s)}{D(s)} = \frac{\frac{1}{s(s+4)}}{1+\frac{G(s)}{s(s+4)}} = \frac{1}{s^2+4s+G(s)}$$

b).

$$s^2+4s+K_p+K_d s = s^2+(4+K_d)s+K_p$$

$$\Rightarrow \begin{cases} 2\zeta\omega_n = 4+K_d \\ \omega_n^2 = K_p \end{cases} \Rightarrow \begin{cases} K_p = \omega_n^2 \\ K_d = 2\zeta\omega_n - 4 \end{cases}$$

$$\Rightarrow \begin{cases} \omega_n = \sqrt{K_p} = 5 \\ 100\% \cdot e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} = 5\% \end{cases}$$

$$\Rightarrow \begin{cases} K_p = 25\ s^{-2} \quad K_d = 2.9\ s^{-1} \\ \zeta = 0.69 \end{cases}$$

c)

$$s^2+4s+K_p+K_d s+\frac{K_i}{s}$$

$$= s^3+(4+K_d)s^2+K_p s+K_i$$

s^3	1	K_p	0
s^2	$4+K_d$	K_i	0
s^1	b_1	0	
s^0	b_2		

$$b_1 = -\frac{1}{4+K_d}\begin{vmatrix} 1 & K_p \\ 4+K_d & K_i \end{vmatrix} =$$

$$= -\frac{1}{4+K_d}\left[K_i - K_p(4+K_d)\right]$$

$$b_2 = -\frac{1}{b_1}\begin{bmatrix} 4+K_d & K_i \\ b_1 & 0 \end{bmatrix}$$

$$= -\frac{1}{b_1}(-K_i b_1)$$

$$= K_i$$

$$\begin{matrix} b_1 > 0 \\ b_2 > 0 \end{matrix} \Rightarrow \begin{cases} -\frac{1}{4+K_d}\left[K_i - K_p(4+K_d)\right] > 0 \\ K_i > 0 \end{cases}$$

$k_i < 25(4+2.9) \approx 172.5 \ s^{-1}$

$0 \sim 172.5 \ s^{-1}$